

Frontend Topics

- Frontend Intro
- Code Editor setup - (VScode)
- Git & Github Basics (For pushing code)

HTML

- HTML Intro
- Elements, Attributes
- Comments
- Inspecting Elements in Browser
- Media types and formats on the web (<audio> and <video>)
- HTML content categories:
 - Metadata content,
 - Flow content,
 - Sectioning content,
 - Heading content,
 - Phrasing content,
 - Embedded content,
 - Interactive content,
 - Palpable content,
 - Form-associated content,
 - Transparent content model
- Inline-level Element and Block-level Elements
- HTML Forms
- What is Semantic Element?
- HTML-5 concepts
- HTML Outro

CSS

- CSS Intro
- CSS Colors
- Inspecting CSS in Browser
- CSS Properties
- CSS selectors
- CSS values and units
 - Functions: calc(), url()
 - Units: px, % (percentage), rem, em, vh, vw
- CSS Box-Model
- Display inline (vs) block
- CSS Flex-box
- CSS Position
- CSS Animations
- External CSS (vs) Internal CSS (vs) Inline CSS
- Responsive Design
 - Responsive images
 - Responsive Table
 - Responsive columns

Media Query

Viewports (Mobile, Tab, Desktop)

Mobile first approach

- CSS Outro
- Practice-1: Creating Layouts:
Columns, Center an element, Sticky Footer, Breadcrumb navigation, List group with badges, Pagination, Card, Header menu)
- Practice-2: Create a Static Webpage

Javascript

- JS intro
- JS Basics
 - JS Outputs:
console.log()
document.write()
alert()
prompt()
innerHTML Property

Syntax:

JavaScript Literals

JavaScript Variables

JavaScript Operators

JavaScript Expressions

JavaScript Keywords

JavaScript Comments

JavaScript Data Types

JavaScript Functions

JavaScript Identifiers

Comments in JS

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- Datatypes in JS
 - Variables - var (vs) let (vs) const
 - Operators in JS:
 - 1) Arithmetic Operators,
 - 2) Assignment Operators,
 - 3) Comparison Operators,
 - 4) Logical Operators,
 - 5) Bitwise Operators,
 - 6) Ternary Operator,
 - 7) Comma Operator,
 - 8) Unary Operators,
 - 9) Relational Operators,
 - 10) BigInt Operators,
 - 11) String Operators,

12) in and instanceof operators in JS

13) Chaining (.) / Optional Chaining Operator (?.)

- Operator precedence in JavaScript
- Statements: (if else, switch, break, continue, return)
- Loop: (for, while, do while, for in, for of)
- Performance & Debugging
- Objects in JS
 - OOPS in JS
 - Creating Object in JS
 - JSON Objects
 - JavaScript Object Reference
- Functions in JS
 - Function Declaration (VS) Function Expression
 - Arrow function
 - Difference Between Function and Method
 - Explain the Different Function States in JavaScript
- JavaScript Functions & Properties
 - JS Function Parameters: (argument, rest parameter, default parameter)
 - JS Function Properties
 - JS in-built Functions
 - Call stack
 - Pure Functions
 - Function Binding (call, apply, bind)
 - Function Generator
- Javascript Arrays
- Javascript Array methods
- More on JS Array
 - Array Constructor
 - Array Properties
 - Static Methods
 - Instance Methods
- Array Practice
- JS String
 - String literal (vs) String Constructor
 - Empty string
 - Multiline string
- Basic Operations on JavaScript Strings:
 - length,
 - Concatenation,
 - Escape Characters,
 - Breaking Long Strings,
 - Find Substring of a String,
 - Convert String to Uppercase and Lowercase,
 - String Search in JavaScript,
 - String Replace in JavaScript,
 - Trimming Whitespace from String,
 - Access Characters from String,

String Comparison in JavaScript,
Passing JavaScript String as Objects

- Numbers, its properties and methods
 - Type Coersion
 - Revisiting some important topics
 - 'this' keyword
 - "useStrict" in JS
 - Type Coersion
 - ES6
 - Hoisting in JS
 - Function Callstack
 - Stack memory (vs) Heap memory
 - setTimeout and setInterval
 - Synchronous (vs) Asynchronous
 - Scope
 - Closure
 - Practice Programs
 - Asynchronous Javascript
 - AJAX
 - Fetch API
 - Async Await
 - Promises
 - Using JSON API - Realtime
 - Client side storage
 - Cookies
 - HTML5 Web storage: (Local Storage, Session Storage)
 - Web SQL
 - IndexedDB
 - Map
 - Set
 - Useful builtin Object methods and properties
 - Date
 - Promise
 - BigInt,
 - Boolean,
 - Proxy/Handler,
 - WeakMap,
 - WeakSet,
 - JSON,
 - Function generator,
 - JS Exercises
 - Practice Projects using - JS, HTML, CSS
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React

- React Intro
- Setup
 - Trying react with CDN
 - Installing react with npm
- Rendering elements
- React DOM (vs) JSX
- Functional Component in React
 - Creating a component
 - Component structure, import, export, parent, child, siblings
 - A glimpse of Functional component (vs) Class component
- Attributes in react element
 - className
 - Using styles
 - Event attributes
- Fragments
- Looping List with map
- Passing data via Props
- Event Handling in React
- Intro to Hooks in React
 - useState Hook
 - useEffect Hook
- Refs
- Lifting State Up
- Conditional rendering
- Trying Tic Tac Toe game
- Deploying React apps
- Working with Forms
- Debounce and Throttle
- Practice project - Forms
- Routing in React
 - Installing React Router
 - Setting up route to pages
 - Types of Routers
- Practice project - Routing
- Context API
- Redux
 - Asynchronous Redux
- Redux Toolkit
- Debugging React Apps
- Practice project - with Redux
- Handling HTTP Request
- Practice project - Handling HTTP
- Keys in React
- Custom Hooks
- React Lifecycle

- Advanced concepts
 - Lazy Loading
 - Higher Order Component
 - Code-splitting
 - Secure your React Apps
- React with Typescript
- Unit Test

- A Glympse of Class Component
- React Outro

- Project in React